

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

08

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

In a bag, there are 7 red, 4 white, 33 blue, and 33 yellow cubes. If one of these cubes is selected at random, what is the probability of selecting a cube that is neither blue nor yellow?

A. $\frac{6}{7}$

B. $\frac{7}{11}$

C. $\frac{1}{3}$

D. $\frac{1}{7}$

In which of the following tables is the relationship between the values of x and their corresponding y -values nonlinear?

A.

x	1	2	3	4
y	8	11	14	17

B.

x	1	2	3	4
y	4	8	12	16

C.

x	1	2	3	4
y	8	13	18	23

D.

x	1	2	3	4
y	6	12	24	48

Data set A: 5, 5, 5, 5, 5, 5, 5, 5, 5
Data set B: 5, 5, 5, 5, 5, 5, 5, 5, 5, 100

Which of the following statements about the means and medians of data set A and data set B is true?

- A. Only the means are different.
- B. Only the medians are different.
- C. Both the means and the medians are different.
- D. Neither the means nor the medians are different.

Q4

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many tablespoons are equivalent to 14 teaspoons? 3 teaspoons = 1 tablespoon

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

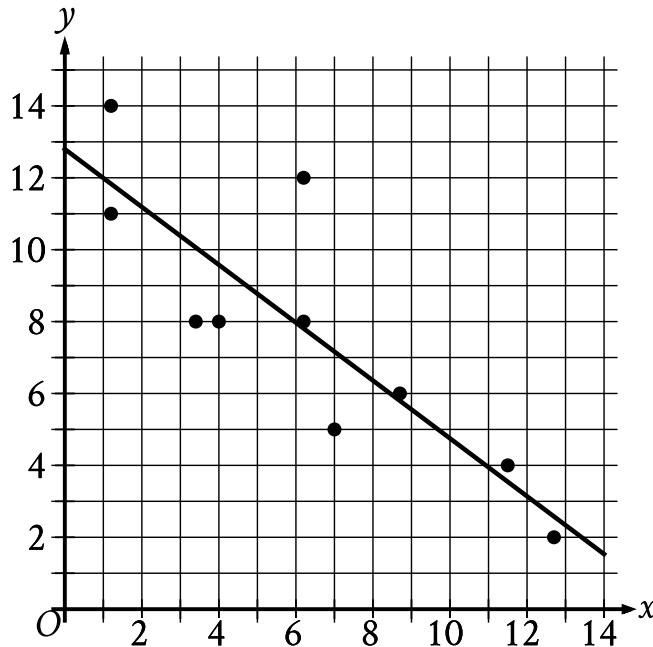
Each vertex of a 14-sided polygon is labeled with one of the 14 letters A through N , with a different letter at each vertex. If one vertex is selected at random, what is the probability that the letter D will be at the selected vertex? (Express your answer as a decimal or fraction, not as a percent.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

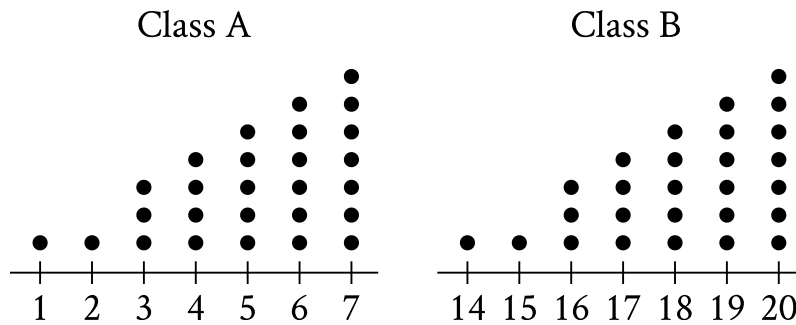
The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
 - The line of best fit slants down from left to right.
 - 2 points are touching the line of best fit.
 - 3 points are above the line of best fit.
 - 5 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 12.9)
 - (6 comma 8)
 - (12 comma 3.2)

Which of the following is closest to the slope of the line of best fit shown?

- A. -2.4
- B. -0.8
- C. 0.8
- D. 2.4



- For the dot plot titled Class A:
 - The number line ranges from 1 to 7 in increments of 1.
 - The data for the dot plot are as follows:
 - 1: 1 dot
 - 2: 1 dot
 - 3: 3 dots
 - 4: 4 dots
 - 5: 5 dots
 - 6: 6 dots
 - 7: 7 dots
- For the dot plot titled Class B:
 - The number line ranges from 14 to 20 in increments of 1.
 - The data for the dot plot are as follows:
 - 14: 1 dot
 - 15: 1 dot
 - 16: 3 dots
 - 17: 4 dots
 - 18: 5 dots
 - 19: 6 dots
 - 20: 7 dots

Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

- A. The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.
- B. The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.
- C. The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.
- D. There is not enough information to compare these standard deviations.

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A distance of 61 furlongs is equivalent to how many feet?

1 furlong = 220 yards and 1 yard = 3 feet

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

Last year, Cedric had 35 plants in his garden. This year, the number of plants in Cedric's garden is 60% greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.